

BRAIN COMMUNICATIONS

Central noradrenergic deficiency in post-infectious chronic fatigue: neurobehavioral correlates

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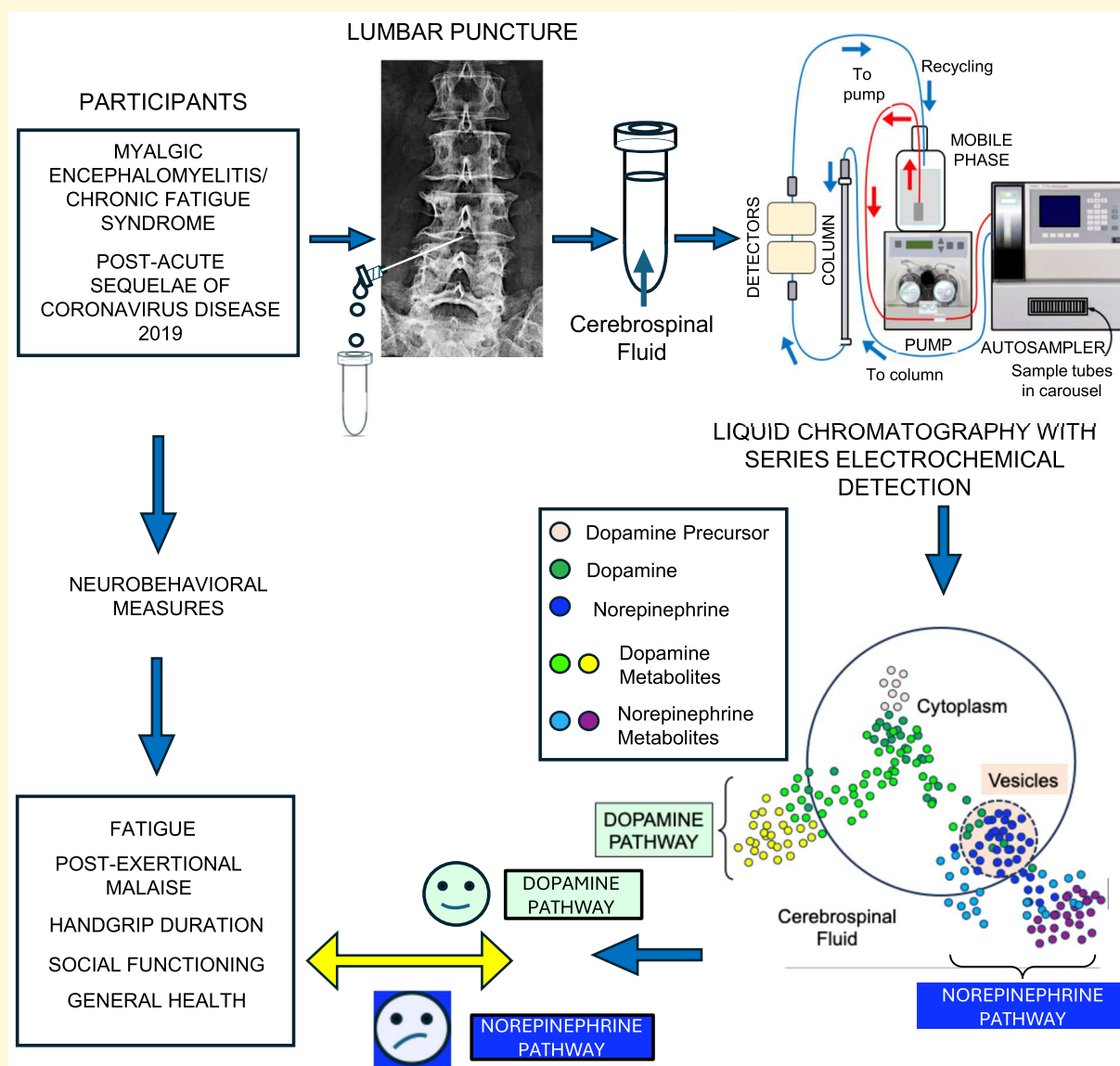
Fatigue, brain fog and post-exertional malaise are major features of post-infectious myalgic encephalomyelitis/chronic fatigue syndrome and post-acute sequelae of severe acute respiratory syndrome coronavirus 2. To date, no specific neurotransmitter abnormalities have been found in either condition. We examined the possibility of central catecholaminergic involvement and clinical correlates in post-infectious myalgic encephalomyelitis/chronic fatigue syndrome and post-acute sequelae of severe acute respiratory syndrome coronavirus 2 groups compared to healthy volunteers and, as positive controls, Parkinson's disease patients. In an observational, cross-sectional cohort study conducted at the National Institutes of Health Clinical Center we measured CSF levels of catecholamines and metabolites in the four groups and assessed correlations with neurobehavioral measures. CSF indices of the central Norepinephrine Pathway (norepinephrine + 3,4-dihydroxyphenylglycol + 3-methoxy-4-hydroxyphenylglycol) and Dopamine Pathway (dopamine + 3,4-dihydroxyphenylacetic acid + homovanillic acid) were measured and related to patient-recorded outcomes and physiological assessments. Mean values for Norepinephrine Pathway activity (in pmol/mL) were lower in the post-infectious myalgic encephalomyelitis/chronic fatigue syndrome, post-acute sequelae of severe acute respiratory syndrome coronavirus 2, and Parkinson's disease groups compared to the healthy volunteer cohort (post-infectious myalgic encephalomyelitis/chronic fatigue syndrome (-15.9 , 95% confidence interval $[-26.1, -5.7]$, $P = 0.0006$); post-acute sequelae of severe acute respiratory syndrome coronavirus 2 (-9.62 , $[-17.9, -1.4]$, $P = 0.015$); Parkinson's disease (-19.4 , $[-27.5, -11.3]$, $P < 0.0001$)). Post-acute sequelae of severe acute respiratory syndrome coronavirus 2 participants with post-exertional malaise had evidence of central noradrenergic deficiency compared to healthy volunteers (-18.3 $[-31.3, -5.3]$, $P = 0.0018$). The post-infectious myalgic encephalomyelitis/chronic fatigue syndrome and post-acute sequelae of severe acute respiratory syndrome coronavirus 2 groups did not differ from the healthy group in values for the Dopamine Pathway index. Across all participants, Norepinephrine Pathway activity correlated positively with handgrip duration and general health and negatively with fatigue. We conclude that post-infectious myalgic encephalomyelitis/chronic fatigue syndrome and post-acute sequelae of severe acute respiratory syndrome coronavirus 2 feature a specific central neurotransmitter pattern involving noradrenergic but not dopaminergic deficiency. The noradrenergic abnormality is associated with major symptoms such as post-exertional malaise.

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Graphical Abstract



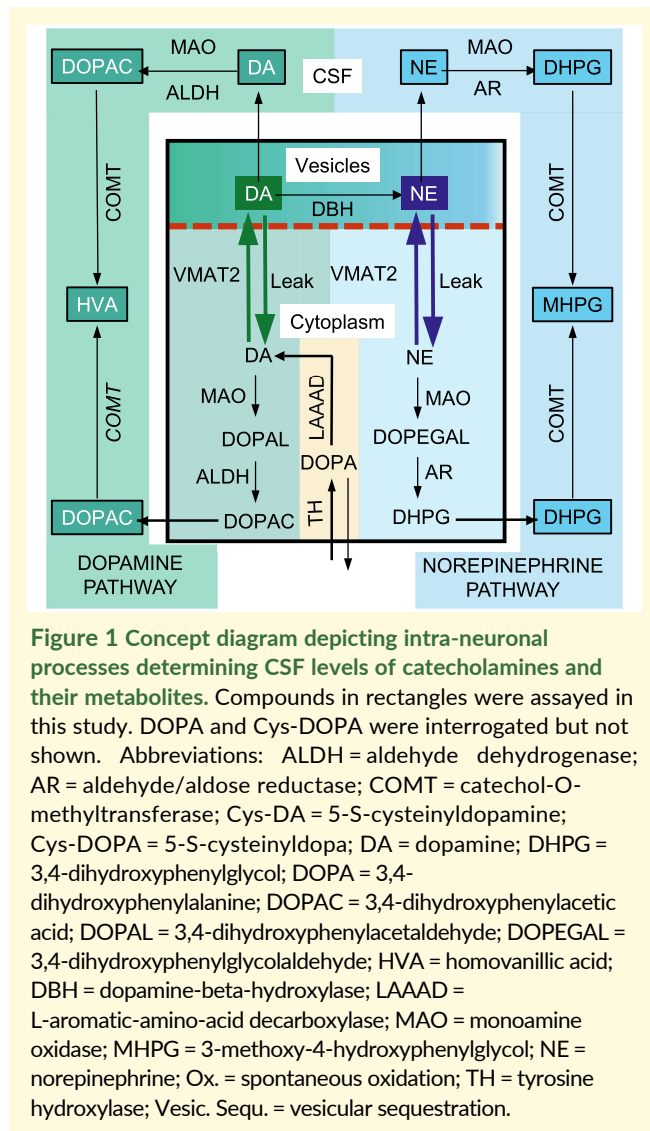
Introduction

Post-Infectious myalgic encephalomyelitis/chronic fatigue syndrome (PI-ME/CFS) and post-acute sequelae of severe acute respiratory syndrome coronavirus 2 (PASC, also known as Long COVID) are significant current public health challenges.^{1,2} Both conditions are defined by specific clinical diagnostic criteria and share several common features. These include a history of prior infection, similar characteristic symptoms, and the absence of a clear mechanistic explanation or definable biomarkers.

PI-ME/CFS is characterized by fatigue, malaise, and cognitive difficulties that worsen significantly after physical or mental exertion, a phenomenon known as post-exertional malaise. PASC patients can also have post-exertional malaise, suggesting a potential common underlying mechanism.

Measurements of CSF concentrations of catecholamines and their metabolites provide information about central dopaminergic and noradrenergic systems.³ In particular, central norepinephrine deficiency is reflected by low CSF levels of 3,4-dihydroxyphenylglycol (DHPG), the main neuronal metabolite of norepinephrine, and of 3-methoxy-4-hydroxyphenylglycol (MHPG), the main end-product of central norepinephrine metabolism; while central dopamine deficiency is reflected by low CSF levels of 3,4-dihydroxyphenylacetic acid (DOPAC), the main neuronal metabolite of dopamine, and of homovanillic acid (HVA), the main end-product of central dopamine metabolism (see Fig. 1 concept diagram).

We recently reported that PI-ME/CFS entails decreased CSF levels of DOPAC and DHPG, as well as of the catecholamine precursor DOPA^{4,5}; however, no previous report has described CSF levels of MHPG or HVA in PI-ME/CFS or PASC.



In this study, we explored whether PI-ME/CFS or PASC cohorts demonstrate common central catecholaminergic abnormalities. Healthy volunteers served as negative controls and Parkinson's disease patients as positive controls, since Parkinson's disease patients exhibit both central noradrenergic and dopaminergic deficiencies.³

To analyse the neurochemical data, we devised, validated, and applied novel indices of central dopaminergic and noradrenergic pathway activities. For the former, we used the sum of DA + DOPAC + HVA, and for the latter, NE + DHPG + MHPG (Fig. 1). If index levels differed as a function of subject group, we analysed correlation matrices of neurochemical data versus patient-reported indices of neurobehavioral and physiological functions and overall health.

One may ask, would it not be sufficient to report on CSF norepinephrine alone as the most direct measure of central noradrenergic activity, since circulating norepinephrine does not pass through the blood-brain barrier (BBB)? Although it is correct that there is an efficient BBB for circulating catecholamines,⁶ this does not mean that CSF norepinephrine straightforwardly reflects 'activity' in the sense of exocytotic release in response to pathway traffic. This is because

most of the released norepinephrine is recycled by reuptake into the nerve terminals and metabolized before the norepinephrine reaches the CSF space. For instance, if there were decreased activity of the cell membrane norepinephrine transporter (NET), CSF norepinephrine would overestimate the rate of exocytosis. Simultaneous measurement of DHPG (the main neuronal metabolite of NE) avoids this sort of misinterpretation. Thus, in humans, treatment with atomoxetine, which attenuates NET activity, increases CSF norepinephrine but also increases the CSF NE/DHPG ratio,⁷ indicating that the increase in CSF norepinephrine is at least partly the result of decreased neuronal reuptake into the cytoplasm and consequently decreased availability of norepinephrine for metabolism by monoamine oxidase to form DHPG. If there were an actual increase in exocytosis, then both norepinephrine and DHPG would be increased in the extracellular fluid.⁸ By the same reasoning, if there were a decrease in exocytosis, then CSF levels of norepinephrine and DHPG would both be decreased.⁹ It is also important to recognize that the main determinant of norepinephrine turnover is not exocytosis (with escape of reuptake) but leakage of vesicular norepinephrine into the cytoplasm.¹⁰ This leakage occurs continuously, relatively independently of exocytosis. This is why, if one graphs CSF DHPG versus norepinephrine, the y-intercept value is above the origin.¹¹ In the present report, by Norepinephrine Pathway 'activity' we do not mean exocytotic release but the turnover of norepinephrine, which under steady-state conditions is balanced by synthesis. CSF norepinephrine, DHPG and MHPG provide related but distinctive and complementary information about the overall status of the central noradrenergic system, just as CSF dopamine, DOPAC, and HVA provide related but distinctive information about the central dopaminergic system.³

As will be seen, in this study, we analysed values for numerous dependent behavioural and neurochemical variables that theoretically are related to each other but also are distinct. This multiplicity enabled reaching strong conclusions by convergence—but at the cost of rather daunting complexity. We wished to summarize the neurochemical information by validating and applying overall indices of central dopaminergic and noradrenergic function that readers could comprehend. Since the chosen indices were novel, we included data from positive controls (a Parkinson's disease group 3) and negative controls (a healthy volunteer group).

Materials and methods

Ethics statement

All the subjects in this study had participated in clinical research protocols approved by the Institutional Review Board (IRB) of the National Institute of Neurological Disorders and Stroke at the National Institutes of Health (NIH) or by the NIH Central IRB (Supplementary Table 1). Written informed consent had been obtained before any procedures were done.

Study design

This was a retrospective analysis of observational data from cohorts of selected patients with PI-ME/CFS, PASC, or Parkinson's disease who had been evaluated at the NIH Clinical Center.

Selection criteria

PI-ME/CFS

Inclusion required participants to have been previously healthy, followed by evidence of infection within the past 5 years and ongoing severe fatigue accompanied by post-exertional malaise. Participants were screened through eligibility interviews and medical record reviews, and their data were included for analysis after an adjudication process. The selection criteria and outcomes for this protocol (NCT02669212) were previously published.¹

PASC

Data from two ongoing protocols (NCT04573062 for PASC, NCT04564287 for PASC with neurological symptoms (Neuro-PASC)) were analysed. The patients were required to have a positive SARS-CoV-2 nucleocapsid test and incomplete recovery with ongoing symptoms after the acute illness. Similar to the PI-ME/CFS group, participants were screened through eligibility interviews and medical record reviews. Adjudication for data inclusion required consensus among three board-certified physicians.

Parkinson's disease

Parkinson's disease patients had been referred for evaluation by the Autonomic Medicine Section of the Division of Intramural Research, National Institute of Neurological Disorders and Stroke (NINDS) under NIH protocols NCT00059033, NCT00775853, NCT03104725, or NCT03648905. There was no recruitment through advertisement. For each participant, medical documentation and a referral letter from their physician were reviewed, eligibility was determined by the Principal Investigator, and the participants were prioritized for inpatient or outpatient evaluation.

Healthy volunteers had been studied as inpatients at the NIH Clinical Center under several intramural NINDS clinical research protocols (Supplementary Table 1).

Lumbar puncture

Participants underwent lumbar puncture either directly or under fluoroscopic guidance. CSF was collected into pre-chilled 1.6 cc plastic tubes, placed immediately in dry ice, and stored at or below -80°C until thawed for assay.³

Norepinephrine and dopamine pathway indices

We simultaneously measured CSF levels of norepinephrine and its metabolites DHPG and MHPG, and of dopamine and its metabolites DOPAC and HVA by liquid chromatography with series post-column electrochemical detection, as described previously by our group.³

We used the sum of CSF norepinephrine + DHPG + MHPG (referred to here as the Norepinephrine Pathway) as an index of central noradrenergic function and CSF dopamine + DOPAC + HVA (referred to as the Dopamine Pathway) as an index of central dopaminergic function (Fig. 1).

Clinical outcome measures

Questionnaires

The Patient-Reported Outcomes Measurement Information System (PROMIS) is a validated assessment tool evaluating various dimensions of health status and well-being (Fatigue, Pain Intensity, Depression, Anxiety, Sleep Disturbances, Pain Interference, Pain Behaviour, Global Health, Sleep-Related Impairments).¹² Test results utilize a T-score metric with a mean of 50 and a standard deviation of 10. Higher scores indicate more of the measured item. PROMIS Global Health is split into Physical and Mental Health components.

The Short Form-36 (SF-36) is a self-reported assessment of quality of life that has been validated across various patient populations.¹³ The SF-36 has eight domains—physical functioning, physical role limitations, social functioning, emotional role limitations, bodily pain, vitality, general health and mental health. These entries are combined into two components, physical and mental. Higher scores indicate fewer health complaints.

The Polysymptomatic Distress (PSD) Scale is a self-administered instrument that assesses the location and severity of pain and tenderness on a person's body and can be used to evaluate fibromyalgia.¹⁴ The PSD provides a total score along with two sub-scores, symptom severity and widespread pain. Higher scores indicate greater reported pain.

The McGill Pain Questionnaire asks patients to pick from various 20-word descriptors of their pain.¹⁵ Higher scores indicate more pain.

The Multiple Ability Self-Report Questionnaire (MASQ) examines five cognitive domains—language, visuo-perceptual, verbal memory, spatial memory, and attention.¹⁶ Higher scores indicate greater cognitive difficulties.

The Multidimensional Fatigue Inventory (MFI) is a self-reported 20-item instrument that measures fatigue across the following subscales—general fatigue, physical fatigue, reduced activity, reduced motivation, and mental fatigue. Higher scores indicate greater fatigue.

The Patient Health Questionnaire-15 (PHQ-15) is a validated self-administered instrument to identify somatic symptoms, symptom severity, and potential somatisation.¹⁷ The PHQ-15 gives a single total score, with a higher score indicating more severe somatic symptoms.

The Pittsburgh Sleep Quality Index (PSQI) is a self-administered questionnaire evaluating sleep quality over a one-month period.¹⁸ Higher scores indicate worse sleep quality.

The Neuropathic Pain Scale (NPS) is a validated self-administered scale used to measure and assess neuropathic pain.¹⁹ Higher scores indicate greater neuropathic pain.

Neuropsychological testing

The following neuropsychological assessments were administered by a trained psychometrist:

The Controlled Oral Word Association Test (COWAT) assesses verbal fluency by cueing patients to spontaneously produce words belonging to a given category or starting with a certain letter.²⁰ Scores are given as T-scores, with a mean of 50 and a standard deviation of 10. Higher scores indicate better performance.

The Brief Visual Memory Test-Revised (BVMT) has participants memorize and reproduce a list of designs.²¹ In the BVMT Delayed Recall (BVMT-DR), participants recall previously memorized designs after a 25-minute delay. Scores are given as T-scores, with a mean of 50 and a standard deviation of 10. Higher scores indicate better performance.

The Hopkins Verbal Learning Test-Revised (HVLRT) has participants memorize and reproduce a list of words.²² In the HVLRT Delayed Recall (HVLRT-DR), the person must recall the list of previously memorized words. Scores are given as T-scores, with a mean of 50 and a standard deviation of 10. Higher scores indicate better performance.

The Effort-Expenditure for Rewards Task (EefRT)²³ examines participant task-related fatigue, effort expenditure, and sensitivity to rewards.

Grip strength

Grip strength, a measure that assesses overall health and fatigability,⁵ was assessed in each hand using a dynamometer. To calculate maximum grip force, participants underwent two trials in which they gripped with maximal possible force for 5 s while seated. Maximum grip force was averaged across both trials. After the maximum grip force was calculated, the participants were instructed to grip the dynamometer again until the force achieved fell below 50% of the maximum grip strength.

Post-exertional malaise qualitative interview

All PASC participants underwent a qualitative interview to assess for post-exertional malaise.²⁴ As required by the most recent diagnostic criteria for ME/CFS,²⁵ all PI-ME/CFS participants were confirmed to have post-exertional malaise as a symptom.

Orthostatic measurements

Orthostatic blood pressure changes were assessed through measurements of systolic and diastolic blood pressure by a brachial cuff with the patient lying supine and then standing for 5 min. Orthostatic hypotension was defined by a decrease in systolic pressure of ≥ 20 mmHg or in diastolic pressure of ≥ 10 mmHg.

The occurrence of excessive orthostatic tachycardia, as is found in postural tachycardia syndrome, was examined using a motorized tilt table in protocols NCT02669212 and NCT0457306. Participants were tilted head-up at a 70-degree angle. An increase in heart rate of ≥ 30 BPM or to ≥ 120 BPM at 10 min of tilting was considered indicative of postural tachycardia syndrome.

Data analysis and statistics

For each subject group, means, standard deviations, medians and ranges of values were calculated for age, body mass index (BMI) and elapsed time from infection. Race, ethnicity and birth sex were summarized by numbers and percentages for the healthy volunteer group and the PI-ME/CFS, PASC, and Parkinson's disease groups. One-way analyses of variance (ANOVAs) were conducted on numerical variables. Significant ANOVAs were then examined using Tukey's post hoc test. Chi-squared tests of independence were run on categorical variables to assess differences in frequencies across groups.

Correlation coefficients were calculated among the neurochemical and neurobehavioral variables in each group using Spearman pair-wise correlations at 95% confidence. Correlations were done separately for the following groups: (1) PI-ME/CFS, PASC and healthy volunteers and (2) PI-ME/CFS and PASC.

Statistical analyses used R version 4.4.1 for Windows (R Foundation for Statistical Computing, Vienna, Austria). Results were given as (estimate [95% confidence interval], *P*-value). No correction for multiple comparisons was applied across the various statistical measurements. There was no sample size calculation, as this was not the primary analysis for the protocols. Figures were generated using the ggplot2 package.²⁶

Results

Participant characteristics

The groups were similar in sex composition and body mass indices (Table 1). The Parkinson's disease group was older than the other groups, with a mean age of 64.1 years (SD 10.6; $\approx < 0.0001$ by Tukey's test), and the healthy volunteer group was older than the PI-ME/CFS group (49.2 versus 38.7 years; $P = 0.027$). The PI-ME/CFS group had roughly double the elapsed time since infection compared to the PASC group (33.2 versus 16.6 months; $P < 0.0001$ by *t*-test).

CSF catecholamines and metabolites levels

Individual values and group mean levels of CSF catecholamines and metabolites by group are shown in Fig. 2.

Central norepinephrine and dopamine pathways

Across the healthy volunteer, PI-ME/CFS, PASC, and Parkinson's disease cohorts, values for indices of the Norepinephrine and Dopamine Pathways varied as a function of group ($F = 14.22$, $df = 3$, $P < 0.0001$; $F = 17.61$, $df = 3$, $P < 0.0001$; Fig. 2A, B). The PI-ME/CFS, PASC, and Parkinson's disease groups had lower mean Norepinephrine Pathway activity than did the healthy volunteer group (PI-ME/CFS: $-15.9 [-26.1, -5.7]$, $P = 0.00055$; PASC: $-9.62 [-17.9, -1.4]$, $P = 0.015$; Parkinson's disease: $-19.4 [-27.5, -11.3]$, $P < 0.0001$; Fig. 2A). The PASC and PI-ME/CFS groups did not differ in Norepinephrine Pathway indices. Mean Norepinephrine Pathway activity in the PI-ME/CFS group did not differ from the Parkinson's disease group. The Parkinson's disease group had lower levels of Norepinephrine Pathway activity than did the PASC group ($P = 0.017$ by Tukey's test).

CSF MHPG levels were lower in the PI-ME/CFS (12.6 [4.6, 20.6], $P = 0.00044$), PASC (7.8 [1.4, 14.3], $P = 0.01035$), and Parkinson's disease (16.0 [9.7, 22.3], $P < 0.0001$) groups compared to the healthy volunteer group.

When the PASC group was divided into two subgroups, one with and one without post-exertional malaise, the subgroup with post-exertional malaise had lower values for Norepinephrine Pathway activity than did the healthy volunteer group (-18.3

Table 1 Participant demographics

	HV (N = 40)	PASC (N = 34)	PD (N = 32)	PI-ME/CFS (N = 16)	P	Overall (N = 122)
Age at Evaluation (years)						
Mean (SD)	49.2 (13.2)	45.2 (12.1)	64.1 (10.6)	38.7 (14.8)	<0.0001	50.6 (15.1)
Median [Min, Max]	53.5 [19.0, 70.1]	44.0 [23.0, 63.0]	66.1 [42.3, 78.2]	39.5 [19.0, 60.0]		54.0 [19.0, 78.2]
Race						
Black	4 (10.0%)	2 (5.9%)	3 (9.4%)	0 (0%)	0.14	9 (7.4%)
Multiracial	3 (7.5%)	0 (0%)	0 (0%)	0 (0%)		3 (2.5%)
White	33 (82.5%)	31 (91.2%)	28 (87.5%)	14 (87.5%)		106 (86.9%)
Asian	0 (0%)	1 (2.9%)	1 (3.1%)	2 (12.5%)		4 (3.3%)
Birth Sex						
Female	13 (32.5%)	23 (67.6%)	12 (37.5%)	9 (56.3%)	0.012	57 (46.7%)
Male	27 (67.5%)	11 (32.4%)	20 (62.5%)	7 (43.8%)		65 (53.3%)
BMI (kg/m ²)						
Mean (SD)	26.6 (3.82)	28.0 (5.58)	26.5 (4.39)	25.9 (5.48)	0.40	26.9 (4.73)
Median [Min, Max]	26.0 [20.6, 35.2]	27.0 [17.6, 47.2]	26.4 [17.1, 34.7]	25.3 [19.4, 37.5]		26.1 [17.1, 47.2]
Elapsed Time Since Infection (months)						
Mean (SD)		16.6 (11.3)		33.2 (15.7)	<0.0001	21.9 (14.9)
Median [Min, Max]		12.5 [1.00, 48.0]		23.5 [13.0, 59.0]		20.0 [1.00, 59.0]

Abbreviations: PD = Parkinson's disease; HV = healthy volunteer; PASC = post-acute SARS-CoV-2; PI-ME/CFS = post-infectious myalgic encephalomyelitis/chronic fatigue syndrome. **Notes:** PD participants were older than other participant groups, with a mean age of 64.1 years (SD 10.6; $P < 0.0001$ by Tukey's test). HVs were older than PI-ME/CFS participants (49.2 versus 38.7 years; $P = 0.027$). PI-ME/CFS participants had roughly double the elapsed time since infection compared to the PASC participants (33.2 versus 16.6 months; $P < 0.0001$ by t -test).

$[-31.3, -5.3]$, $P = 0.0018$ by Tukey's test; Fig. 2C), whereas the subgroup without post-exertional malaise did not.

The Parkinson's disease group had a lower mean value for Dopamine Pathway activity than did the other groups (healthy volunteer and PASC: $P < 0.0001$, PI-ME/CFS: $P = 0.0044$). The PI-ME/CFS and PASC groups did not differ from the healthy volunteer group in values for the Dopamine Pathway index.

Although the healthy volunteer and patient groups differed in sex makeup, analysis of covariance showed that sex did not influence the difference between the healthy volunteer and patient groups in values for the Norepinephrine Pathway index (data not shown).

Clinical correlates of norepinephrine pathway activity

All participants

Across all subjects ($N = 63$), individual values for Norepinephrine Pathway activity correlated positively with handgrip duration (right hand: $\rho = 0.54$ [0.32, 0.70], $P < 0.0001$, left hand: $\rho = 0.40$ [0.15, 0.60], $P = 0.0027$; Supplementary Table 2). Norepinephrine Pathway activity also correlated positively with PROMIS mental and physical health scores ($\rho = 0.33$; $\rho = 0.38$) and physical functioning, physical role limitations, social functioning, vitality, general health, and mental and physical component sub-scores of the SF-36 (correlation coefficients 0.32–0.45, Supplementary Table 2).

Statistically significant negative correlations between values for the Norepinephrine Pathway index and neurobehavioral measures are also shown in Supplementary Table 2. The main negative correlations were for the PSD scale, some PROMIS measures, and the MFI.

PI-ME/CFS and PASC

Among the PI-ME/CFS and PASC patients considered as a single group ($n = 38$), Norepinephrine Pathway activity correlated positively with handgrip duration (right hand: $\rho = 0.62$ [0.35, 0.80], $P = 0.00011$; left hand: $\rho = 0.48$ [0.15, 0.71], $P = 0.0058$; Table 2). Norepinephrine Pathway activity also correlated positively with the social functioning, vitality, and general health sub-scores of the SF-36, with correlation coefficients ranging from 0.36 to 0.45 (Table 2).

Values for the Norepinephrine Pathway index correlated negatively with general fatigue and mental fatigue sub-scores of the MFI, the emotional role limitations sub-score of the SF-36, and the PROMIS fatigue score, with correlation coefficients ranging from -0.43 to -0.36 (Table 2).

Orthostatic measurements

Among the PI-ME/CFS and PASC patients considered as a single group, Norepinephrine Pathway index values were unrelated to the frequencies of orthostatic hypotension ($P = 0.56$) or excessive orthostatic tachycardia ($P = 0.35$). Within the PI-ME/CFS and PASC groups, there were too few instances of orthostatic hypotension or excessive orthostatic tachycardia to conduct meaningful calculations of correlation coefficients.

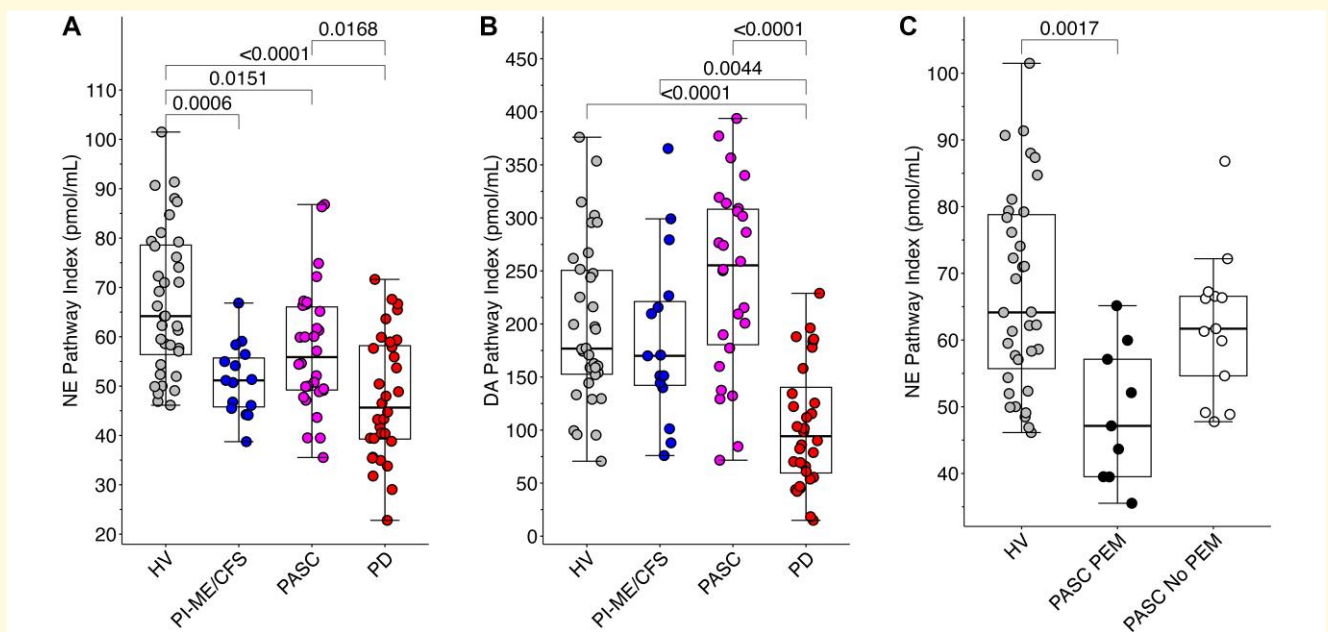


Figure 2 Dot-and-whisker plots for central noradrenergic and dopaminergic indices across subject groups. In A-B, compared groups are healthy volunteers (HV), post-infectious myalgic encephalomyelitis/chronic fatigue syndrome (PI-ME/CFS), post-acute sequelae of COVID-19 with neurological symptoms (PASC), and Parkinson's disease (PD). Numbers are *P*-values for comparisons between groups by Tukey's post-hoc test following one-way factorial analyses of variance. (A) Plot depicting CSF Norepinephrine Pathway in pmol/mL for healthy volunteers (*n* = 36), participants with PI-ME/CFS (*n* = 15), participants with PASC (*n* = 30), and participants with PD (*n* = 32). (B) Plot depicting CSF Dopamine Pathway in pmol/mL for healthy volunteers (*n* = 34), participants with PI-ME/CFS (*n* = 15), participants with PASC (*n* = 26), and participants with PD (*n* = 32). In C, compared groups are healthy volunteers (HV) (*n* = 35), post-acute sequelae of COVID-19 with neurological symptoms and reported post-exertional malaise (PASC post-exertional malaise) (*n* = 9), and post-acute sequelae of COVID-19 with neurological symptoms and no reported post-exertional malaise (PASC No post-exertional malaise) (*n* = 13).

Effects of medications

We reviewed all the medications that patients were on and identified the participants who were on a 'Query' medication at the time the CSF was obtained. By Query we mean a tricyclic of any sort (including cyclobenzaprine, the chemical structure of which is tricyclic), an amphetamine, venlafaxine, or duloxetine. There were 10 patients on a Query drug (Supplementary File 1). The Query subgroup had higher NE/DHPG ratios than did the non-Query subgroup and healthy volunteers (Supplementary Fig. 1). Both the Query subgroup and the non-Query subgroup had lower mean values for the index of central Norepinephrine Pathway activity (CSF NE + DHPG + MHPG) than did the healthy volunteers.

Discussion

The present data indicate that PI-ME/CFS and PASC have in common a central neurochemical abnormality—noradrenergic deficiency. This abnormality seems to be neurotransmitter-specific, since we found no evidence for central dopaminergic deficiency in either condition. Low activity of the Norepinephrine Pathway was associated with decreased scores on general health indices, such as physical and mental functioning, vitality and with decreased motor

scores such as shortened handgrip duration. In the PI-ME/CFS and PASC groups, Norepinephrine Pathway activity did not correlate with other symptoms such as pain, altered cognition, decreased memory, anxiety, or depression. Within the PASC group, Norepinephrine Pathway activity was associated with post-exertional malaise.

MHPG is the main end-product of norepinephrine metabolism in the brain and is the predominant analyte in the index of Norepinephrine Pathway activity, CSF norepinephrine + DHPG + MHPG. We observed decreased CSF MHPG levels in both PI-ME/CFS and PASC compared to healthy volunteers. Low CSF MHPG levels have been reported in patients with pure autonomic failure²⁷ or multiple system atrophy,²⁸ but CSF MHPG has not been reported previously in PI-ME/CFS or PASC.

Various forms of autonomic dysfunction have been noted in PI-ME/CFS²⁹⁻³¹ and PASC;³²⁻³⁴ however, we found low frequencies of both orthostatic hypotension and excessive orthostatic tachycardia in these cohorts considered as a single group.

What could explain central noradrenergic but not dopaminergic deficiency in PI-ME/CFS and PASC? Norepinephrine synthesis requires vesicular uptake of dopamine from the cytoplasm, because dopamine-beta-hydroxylase, the enzyme that catalyses the conversion of dopamine to norepinephrine, is localized to the vesicles (Fig. 1), and vesicular uptake and retention of catecholamines rely heavily on an ATP-dependent proton pump. In contrast, steps in

Table 2 Correlations of the norepinephrine pathway in all patients

Norepinephrine Pathway Correlations				
All Patients (n = 38)				
	Lower CI	r	Upper CI	P
50% Threshold Handgrip Duration				
Right Hand	0.35	0.62	0.80	0.00011
Left Hand	0.15	0.48	0.71	0.0058
36 Item Short-Form Survey				
General Health	0.03	0.36	0.62	0.033
Vitality	0.03	0.36	0.62	0.032
Social Functioning	0.13	0.45	0.68	0.0073
Emotional Role Limitations	−0.62	−0.36	−0.04	0.031
Patient-Reported Outcomes Measurement Information System				
Fatigue	−0.66	−0.42	−0.10	0.011
Multidimensional Fatigue Inventory				
General Fatigue	−0.66	−0.42	−0.11	0.010
Physical Fatigue	−0.66	−0.43	−0.11	0.0094

Abbreviations: DHPG = 3,4-dihydroxyphenylglycol; MHPG = 3-methoxy-4-hydroxyphenylglycol; NE = norepinephrine. **Notes:** Norepinephrine Pathway = CSF NE + CSF DHPG + CSF MHPG (in pmol/mL).

the synthesis and metabolism of cytoplasmic dopamine do not require ATP. Based on these differences, we speculate that the decreased Norepinephrine Pathway activity observed in PI-ME/CFS and in PASC with post-exertional malaise reflects decreased ATP availability for vesicular uptake and retention. Decreased ATP availability would also provide a unifying explanation for clinical manifestations such as fatigue, decreased handgrip duration, and decreased overall sense of health.

When we compared the subgroups of patients who were on or were not on a Query drug at the time of CSF sampling, the Query drug subgroup had higher NE/DHPG ratios than did the non-Query subgroup and the healthy volunteer group, as one would expect from the drugs inhibiting neuronal reuptake of norepinephrine.⁷ In both patient subgroups, the mean values for the Norepinephrine Pathway index were lower than in the healthy volunteer group, meaning that the Norepinephrine Pathway abnormality remained after removing from consideration the data from the Query drug subgroup.

Limitations

This study involved relatively small cohorts studied in a highly comprehensive manner at a single centre. Generalisability of the results to the broader populations of PI-ME/CFS and PASC patients is unknown.

CSF levels of catecholamines and their metabolites vary extensively across individuals. In this study, the CSF samples were collected and processed immediately in a standardized way, minimising variability.

While the PI-ME/CFS cohort was very well defined,⁴ eligibility of the PASC participants was based on neurologic symptoms, and symptomatic assessments are subject to biases. To minimize

influences of such biases, personnel assaying and analysing the CSF data were strictly blinded as to the symptomatic indices until after data tabulation.

There is no clear standard of measurement for post-exertional malaise. We assessed post-exertional malaise using a validated semi-structured qualitative interview that queried symptoms at baseline and after a cardiac stress test.²⁴

Association does not imply causation, and the present results should not be taken to imply that central noradrenergic deficiency is causally related to the neurobehavioral outcome measures. Evaluating causation could be done by examining the effects of manipulations of central norepinephrine release, reuptake, or metabolism on neurobehavioral measures.

We did not systematically control the timing of CSF collection. Our group reported previously on the circadian rhythm of CSF NE.³⁵ During waking hours, there is relatively little change. Moreover, the central noradrenergic index we used was dominated by CSF MHPG, and there is no diurnal pattern of CSF MHPG levels.³⁶

Summary and implications

The results of this study provide support for a specific neurotransmitter abnormality, central noradrenergic deficiency, as a shared feature of PI-ME/CFS and PASC. In these disorders, values for a measure of central noradrenergic deficiency (CSF norepinephrine + DHPG + MHPG, referred to as the Norepinephrine Pathway) were related to several patient-reported indices of overall health, the duration of handgrip, and post-exertional malaise. Our findings have implications for potential treatments of these forms of infection-associated chronic illness, depending on whether future research confirms that central noradrenergic deficiency in these conditions is pathophysiologically significant.

Supplementary material

Supplementary material is available at *Brain Communications* online.

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Competing interests

Dr. Goldstein retired from the NIH in August, 2025. After retirement, he founded The Autonomic and Catecholamine Healthspan Institute, LLC (abbreviated ACHI). We do not view this as a potential conflict of interest. He also recently joined the Medical Advisory Board of The Dysautonomia Project. We do not view this as a potential conflict of interest.

Data availability

Individual data referred to in the Results, Figures, and Tables are in the [Supplementary Workbook](#) and [4 Supplementary Tables](#). Coded individual data from neurobehavioral questionnaires will be made available by co-author Brian Walitt (email brian.walitt@nih.gov) upon reasonable request. The codes generated and used in this work are available at the repository <https://github.com/lillianaregawi/Central-NE-Pathway-Index-Scripts/>.

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